

# Notes

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## Combinatorics

### Permutations

Sequences without repetition of  $n$ , written  ${}_nP_k$ , are ordered arrangements of  $k$  elements in an  $n$  elements set. The formula for the number of permutations is:

$$\frac{n!}{(n-k)!}$$

### Combination

A *combination* is a selection of elements of a set that has distinct members, such that the order of the selection does not matter.

$$\binom{n}{k}$$

or

$${}_nC_k$$

The number of  $k$ -combinations of a set of cardinality  $n$  is given by:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

### Exercises

1. A club has 10 female and 8 male members. Three members are to be chosen at random to represent the club. What is the probability that 2 women and one man will be chosen? Round your answer to the nearest percent.

First let's figure out how many ways to pick there are in total. The order doesn't matter, so we use a permutation:

$$N = {}_3C_{18} = \frac{18!}{(18-3)!3!} = 816$$

And how many ways to pick one male ( $m$ ) or two women ( $w$ ):

$$m = {}_1C_8 = \frac{8!}{(8-1)!} = 8$$

$$w = {}_2C_{10} = \frac{10!}{(10-2)!} = 45$$

Multiplying these two, we get the numbers of ways to pick one man and two women.

Our final result is:

$$\frac{mw}{N} = \frac{8 \cdot 45}{816} = 0.4411764...$$

**Exercise** What is the probability that among 4 randomly chosen people whose birthdays are all in the last week of September, at least two will have birthdays on the same day? Round your answer to 3 decimal places.

It's easier to answer the question "what is the probability that all their birthdays are on a different day?" and take the complement.

We have 7 days to choose from, so there are  ${}_7P_4 = 840$  permutations. There are  $7^4$  possibilities for all the birthday combinations. We have:

$$P(A') = \frac{n}{N} = \frac{840}{7^4} = 0.3498... P(A) = 1 - P(A') = 0.650...$$

## Geometry

### Circles

**Circumference:**  $2\pi r$

**Surface:**  $\pi r^2$

### Cylinders

**Surface:**  $2\pi r^2 + 2\pi r h$

**Volume:**  $V = \pi r^2 h$

#### Exercises

- Find an expression for the radius  $r$  of the base of a cylinder in terms of its volume  $V$  given that the ratio of the radius to the height is  $3\pi$ .

We have  $\frac{r}{3\pi} = h$ . Let's substitute that in the volume formula:

$$\begin{aligned}\pi r^2 h &= V \\ \pi r^2 \frac{r}{3\pi} &= V \\ \frac{\pi r^3}{3} \pi &= V \\ \frac{r^3}{3} &= V \\ r^3 &= 3V \\ r &= \sqrt[3]{3V}\end{aligned}$$

- Calculate the circumference of the base, given volume is  $625\pi m^2$  and the ratio of its height to its radius is 5.

$$\begin{aligned}\pi r^2 h &= V \\ \pi r^2 \cdot (5r) &= 625\pi \\ 5r^3 &= 625 \\ r^3 &= 125 \\ r &= 5\end{aligned}$$

Now the circumference is  $2\pi r = 2\pi(5) = 10\pi$

### Spheres

**Surface:**  $4\pi r^2$  **Volume:**  $\frac{4}{3}\pi r^3$

### Triangles

#### Surface

The formula for the area of a triangle from one angle  $C$  and its two adjacent sides  $a$  and  $b$  is:

$$\frac{1}{2}ab \sin C$$

$C$  is called the **included angle**.

### Polygons

The sum of the interior angles of a polygon is always  $180 * (n - 2)$  where  $n$  is the number of sides.

## Regular polygons

With  $n$  as the number of sides, exterior angles are given by:

$$\theta = \frac{360}{n}$$

## Logarithm

### Domain

$(0, \infty)$

### Range

$(-\infty, \infty)$

## Falling objects

The *displacement* equation:

$$s = ut + \frac{1}{2}gt^2$$

where:

- $s$  is the **distance** travelled in meters
- $u$  is the **initial velocity** in m/s
- $t$  is the **time** in seconds
- $g$  is the **acceleration due to gravity**, constant equal to  $9.8\text{m/s}^2$

## Geometric Sequences

Geometric sequences are of the form  $a_n = a_1 \cdot r^{n-1}$ , where  $r$  is called the *common ratio*.

### Exercises

1. Given that a geometric sequence has the first term  $a_1 = -2$  and the 4th term  $a_4 = -54$ , what is the index of the term  $-6$ ?

The formula for  $a_n$  is  $a_n = b \cdot r^{n-1}$ . Since  $a_1 = b$ , we have:

$$\begin{aligned}a_4 &= -2 \cdot r^3 \\-54 &= -2 \cdot r^3 \\27 &= r^3 \\r &= \sqrt[3]{27} \\r &= 3\end{aligned}$$

So the formula is  $a_n = -2 \cdot 3^{n-1}$ . Now we are looking for the index. We substitute  $-6$  for  $a_n$ :

$$\begin{aligned}-6 &= -2 \cdot 3^{n-1} \\3 &= 3^{n-1} \\n &= 2\end{aligned}$$

2. First term:  $a_1 = 81$ , common ratio  $r = \frac{1}{3}$ , and  $a_n = 1$ . Find  $n$ .

$$\begin{aligned}
 1 &= 81 \cdot \left(\frac{1}{3}\right)^{n-1} \\
 \frac{1}{81} &= \left(\frac{1}{3}\right)^{n-1} \\
 \left(\frac{1}{3}\right)^4 &= \left(\frac{1}{3}\right)^{n-1} \\
 4 &= n - 1 \\
 n &= 5
 \end{aligned}$$

3. Given  $a_3 = 36$ ,  $r = 3$ , what is the index of the term 108?

First let's find the formula:

$$\begin{aligned}
 36 &= a_1 \cdot 3^{3-1} \\
 36 &= a_1 \cdot 9 \\
 a_1 &= 4
 \end{aligned}$$

So  $a_n = 4 \cdot 3^{n-1}$ . Now,

$$\begin{aligned}
 108 &= 4 \cdot 3^{n-1} \\
 27 &= 3^{n-1} \\
 3^3 &= 3^{n-1} \\
 3 &= n - 1 \\
 n &= 4
 \end{aligned}$$

# Polynomials

## Forms

- Standard form:  $y = ax^2 + bx + c$
- Vertex form:  $y = a(x - h)^2 + k$ 
  - In this form, the coordinates of the vertex are  $(h, k)$
- Factored form:  $y = a(x - x_1)(x - x_2)$

## Vertex

If the binomial is given in standard form, the vertex'  $x$  coordinate is given by  $-\frac{b}{2a}$ . Once you have the  $x$  coordinate, substitute in the equation to find  $y$ .

## Domain and range

The **domain** is always  $(-\infty, +\infty)$ .

The **range** is, for upward opening parabolas, assuming vertex form:  $y = a(x - h)^2 + k$ , the range is  $[k, +\infty)$ .

## Factor theorem

**Theorem.**  $(x - r)$  is a factor of a polynomial  $p(x)$  if and only if the root  $x = r$  is also a root of  $p(x)$ .

## Exercises

- The polynomial  $-5x^4 + 8x^3 + 29x^2 - 20x - 12$  can be expressed as  $-(x - 1)(x - 3) \cdot P(x)$ . Find  $P(x)$ .

First let's divide by  $(x - 1)$

$$\begin{array}{r|rrrrr} 1 & -5 & 8 & 29 & -20 & -12 \\ & & -5 & 3 & 32 & 12 \\ \hline & -5 & 3 & 32 & 12 & 0 \end{array}$$

So we have  $(x - 1) \cdot (-5x^3 + 3x^2 + 32x + 12)$

Let's divide by  $(x - 3)$

$$\begin{array}{r|rrrr} 3 & -5 & 3 & 32 & 12 \\ & & -15 & -36 & -12 \\ \hline & -5 & -12 & -4 & 0 \end{array}$$

So we have  $(x - 1)(x - 3)(-5x^2 - 12x - 4)$

So we have  $(x - 1) \cdot (x - 3) \cdot (-5x^2 - 2x + 4)$

Finally,  $-(x - 1)(x - 3)(5x^2 + 2x + 4)$ . So  $P(x) = 5x^2 + 2x + 4$ .

Factoring further,  $P(x) = (5x + 2)(x + 2)$ .

- Find the polynomial  $P(x)$  given that

$$4x^4 - 19x^3 - 9x^2 + 19x + 5 = (x - 1)(x - 5)P(x)$$

First let's divide by  $(x - 1)$ .

$$\begin{array}{r|rrrrr} 1 & 4 & -19 & -9 & 19 & 5 \\ & & 4 & -15 & -24 & -5 \\ \hline & 4 & -15 & -24 & -5 & 0 \end{array}$$

So  $(x - 1)(4x^3 - 15x^2 - 24x - 5)$ . Now divide by  $(x - 5)$ .

$$\begin{array}{r|rrrr}
 5 & 4 & -15 & -24 & -5 \\
 & & 20 & 25 & 5 \\
 \hline
 & 4 & 5 & 1 & 0
 \end{array}$$

So  $(x - 1)(x - 5)(4x^2 + 5x + 1)$ .

- Find  $P(x)$  given  $4x^4 - 8x^3 - 3x^2 + 7x - 2 = (x - 2)(x + 1)P(x)$ .

First divide by  $(x + 1)$ :

$$\begin{array}{r|rrrrr}
 -1 & 4 & -8 & -3 & 7 & -2 \\
 & & -4 & 12 & -9 & 2 \\
 \hline
 & 4 & -12 & 9 & -2 & 0
 \end{array}$$

So  $(x + 1)(4x^3 - 12x^2 + 9x - 2)$ . Now divide the remainder by  $(x - 2)$ :

$$\begin{array}{r|rrrr}
 2 & 4 & -12 & 9 & -2 \\
 & & 8 & -8 & 2 \\
 \hline
 & 4 & -4 & 1 & 0
 \end{array}$$

So  $(x + 1)(x - 2)(4x^2 - 4x + 1)$ . So  $P(x) = (4x^2 - 4x + 1)$ . Factoring further,

$$\begin{aligned}
 P(x) &= 4x^2 - 4x + 1 \\
 &= (4x^2 - 2x) + (-2x + 1) \\
 &= 2x(2x - 1) - 1(2x - 1) \\
 &= (2x - 1)^2
 \end{aligned}$$

- Let  $p(x) = -6x^4 + 5x^3 + 52x^2 - 19x - 60$ . Given that  $p(-1) = p(3) = 0$ , find the sum of all distinct roots of  $p(x)$ .

$p(-1) = 0$  means  $x = -1$  is a root, so  $(x + 1)$  is a factor.

$p(3) = 0$  means  $x = 3$  is a root, so  $(x - 3)$  is a factor.

Let's first divide by  $(x + 1)$ .

$$\begin{array}{r|rrrrr}
 -1 & -6 & 5 & 52 & -19 & -60 \\
 & & 6 & -11 & -41 & 60 \\
 \hline
 & -6 & 11 & 41 & -60 & 0
 \end{array}$$

So  $p(x) = (x + 1)(-6x^3 + 11x^2 + 41x - 60)$ . Now let's divide the remainder by  $(x - 3)$ .

$$\begin{array}{r|rrrr}
 3 & -6 & 11 & 41 & -60 \\
 & & -18 & -21 & 60 \\
 \hline
 & -6 & -7 & 20 & 0
 \end{array}$$

So  $p(x) = (x + 1)(x - 3)(-6x^2 - 7x + 20)$ .

Factoring the remainder further:

$$\begin{aligned}
 p(x) &= (x + 1)(x - 3)(-6x^2 - 7x + 20) \\
 &= (x + 1)(x - 3)(-6x^2 + 8x) + (-15x + 20) \\
 &= (x + 1)(x - 3)(2x + 5)(-3x + 4)
 \end{aligned}$$

The roots are  $x = -1, 3, -\frac{5}{2}, \frac{4}{3}$ .

The number we are looking for is  $-1 + 3 - \frac{5}{2} + \frac{4}{3} = 2 - \frac{5}{2} + \frac{4}{3} = -\frac{1}{2} + \frac{4}{3} = \frac{5}{6}$ .

- Find the distinct roots of  $x^4 - 5x^2 + 4$  given  $p(1) = p(-1) = 0$ .

$(x - 1)$  is a factor, let's divide by it first:

$$\begin{array}{r|rrrrrr}
 1 & 1 & 0 & -5 & 0 & 4 \\
 & & 1 & 1 & -4 & -4 \\
 & 1 & 1 & -4 & -4 & 0
 \end{array}$$

So  $(x-1)(x^3 + x^2 - 4x - 4)$ . Now divide by  $(x+1)$ .

$$\begin{array}{r|rrrr}
 -1 & 1 & 1 & -4 & -4 \\
 & & -1 & 0 & 4 \\
 & 1 & 0 & -4 & 0
 \end{array}$$

So  $(x-1)(x+1)(x^2-4) = (x-1)(x+1)(x+2)(x-2)$ .

- Find  $P(x)$  given  $x^4 - 2x^3 - 7x^2 + 18x - 18 = (x-3)(x+3)P(x)$ .

First let's divide by  $(x-3)$ :

$$\begin{array}{r|rrrrrr}
 3 & 1 & -2 & -7 & 18 & -18 \\
 & & 3 & 3 & -12 & 18 \\
 & 1 & 1 & -4 & 6 & 0
 \end{array}$$

So the remainder is  $x^3 + x^2 - 4x + 6$ . Let's divide by  $(x+3)$ .

$$\begin{array}{r|rrrr}
 -3 & 1 & 1 & -4 & 6 \\
 & & -3 & 6 & -6 \\
 & 1 & -2 & 2 & 0
 \end{array}$$

We have  $P(x) = x^2 - 2x + 2$ .

- Find  $P(x)$  given that  $x^4 + x^3 - 36x^2 + 37x - 35 = (x-5)(x+7)P(x)$ .

First divide by  $(x-5)$ :

$$\begin{array}{r|rrrrr}
 5 & 1 & 1 & -36 & 37 & -35 \\
 & & 5 & 30 & -30 & 35 \\
 & 1 & 6 & -6 & 7 & 0
 \end{array}$$

Remainder:  $x^3 + 6x^2 - 6x + 7$ . Now divide that by  $(x+7)$ :

$$\begin{array}{r|rrrr}
 -7 & 1 & 6 & -6 & 7 \\
 & & -7 & 7 & -7 \\
 & 1 & -1 & 1 & 0
 \end{array}$$

Result:  $x^2 - x + 1$

## Polynomial long division

### Exercises

- What is the remainder of  $3x^2 - 2x^2 - 18x - 1$  divided by  $3x^2 - 8x - 2$ ?

$$\begin{array}{r}
 \begin{array}{r}
 3x^2 - 8x - 2 \mid 3x^2 - 2x - 18 - 1 \\
 \underline{3x^2 - 8x - 2} \phantom{-1} \\
 6 \phantom{-16} - 16 \phantom{-4} \\
 \underline{6x - 16} \phantom{-4} \\
 3
 \end{array}
 \end{array}$$

The remainder is 3.

- What is the remainder when  $x^4 - 2x^3 - 4x^2 - 10x - 7$  is divided by  $x^2 + 2x + 1$ ?

$$\begin{array}{r}
 \begin{array}{r}
 x^2 - 4x + 3 \mid x^4 - 2x^3 - 4x^2 - 10x - 7 \\
 \underline{x^2 + 2x + 1} \phantom{-7} \\
 -4x^3 - 6x^2 - 10x - 7
 \end{array}
 \end{array}$$



$$\begin{array}{r}
 1 \quad 2 \quad 1 \\
 \hline
 -4 \quad -5 \\
 -4 \quad -8 \quad -4 \\
 \hline
 3 \quad -6 \\
 3 \quad 6 \quad 3 \\
 \hline
 -12x \quad -10
 \end{array}$$

The remainder is  $-12x - 10$ .

- What is the quotient of  $2x^4 - x^2 - 1$  divided by  $2x^2 + 1$ ?

$$\begin{array}{r}
 x^2 \quad -1 \\
 2x^2 + 1 \mid 2 \quad 0 \quad -1 \quad 0 \quad -1 \\
 2 \quad 0 \quad 1 \\
 \hline
 -2 \\
 -2 \quad 0 \quad 1 \\
 \hline
 0
 \end{array}$$

So it's  $x^2 - 1$ .

## Rational roots theorem

The rational roots theorem (a.k.a. rational zeros theorem) gives the rational roots of a polynomial with integer coefficients.

**Theorem.** Suppose that  $\frac{p}{q} \in \mathbb{Q}$ , in lowest terms, is a root of a polynomial with integer coefficients. Then  $p$  must be a factor of the polynomial's constant term, and  $q$  must be a factor of its leading coefficient.

That means every root must take the form  $\pm \frac{\text{factor of the constant term}}{\text{factor of the leading coefficient}}$ .

## Exercises

- For the polynomial  $f(x) = 3x^8 - 18x^2 - 33x - 18$ , how many potential roots are given by the rational roots theorem?

The numerator must be a factor of  $-18$ . The denominator must be a factor of 3. So here are the possibilities:

$$\pm \frac{1}{3}, \pm \frac{2}{3}, \pm 1, \pm 2, \pm 3, \pm 6, \pm 9, \pm 18$$

So 16.

- Given that the polynomial  $2x^3 + x^2 + 5x - 3$  has a root in the interval  $(0, \frac{3}{2})$ , find the sum of the distinct real roots of  $f(x)$ .

Using the rational roots theorem, we can find the potential roots of the polynomial. The numerator must be a factor of  $-3$ , and the denominator must be a factor of 2. So here are the possibilities in the interval  $(0, \frac{3}{2})$ :

$$\frac{1}{2}, 1$$

We can quickly find which one of these roots by evaluating the polynomial at these points. It's  $\frac{1}{2}$ .

Now let's use synthetic division to find the remaining roots:

$$\begin{array}{r|rrrr} 1/2 & 2 & 1 & 5 & -3 \\ & 2 & 2 & 6 & 0 \end{array}$$

So  $2x^3 + x^2 + 5x - 3 = \left(x - \frac{1}{2}\right)(2x^2 + 2x + 6)$ . Since the discriminant of  $2x^2 + 2x + 6$  is negative, it has no real roots. Therefore, the only real root is  $x = \frac{1}{2}$ .

## Rationalizing the denominator

In general, you want to avoid radicals in the denominator. To rationalize the denominator, multiply both the numerator and the denominator by the conjugate of the denominator.

$$\begin{aligned} & \frac{1}{1 + \sqrt{2}} \\ &= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\ &= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\ &= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\ &= \frac{1 - \sqrt{2}}{1 - 2} \\ &= \frac{1 - \sqrt{2}}{-1} \\ &= -1 + \sqrt{2} \end{aligned}$$

## Exercises

•

$$\begin{aligned} & \frac{1}{1 + \sqrt{2}} \\ &= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\ &= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\ &= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\ &= \frac{1 - \sqrt{2}}{1 - 2} \\ &= \frac{1 - \sqrt{2}}{-1} \\ &= -1 + \sqrt{2} \end{aligned}$$

1.

$$\begin{aligned}
& \frac{1 + \sqrt{8}}{\sqrt{9} - \sqrt{2}} \\
&= \frac{(1 + \sqrt{8})(\sqrt{9} + \sqrt{2})}{7} \\
&= \frac{\sqrt{9} + 3\sqrt{8} + \sqrt{2} + \sqrt{16}}{7} \\
&= \frac{3 + 6\sqrt{2} + \sqrt{2} + 4}{7} \\
&= \frac{7 + 7\sqrt{2}}{7} \\
&= 1 + \sqrt{2}
\end{aligned}$$

2.

$$\begin{aligned}
\frac{x^2 y^2}{\sqrt{x^3 y^3}} &= \frac{x^2 y^2 \sqrt{x^3 y^3}}{x^3 y^3} \\
&= \frac{x^2 y^2 x^{3/2} y^{3/2}}{x^3 y^3} \\
&= \frac{x^{7/2} y^{7/2}}{x^3 y^3} \\
&= \sqrt{xy}
\end{aligned}$$

3.

$$\begin{aligned}
\frac{\sqrt{12x}}{3\sqrt{2x^2 y}} &= \frac{\sqrt{12x} \cdot \sqrt{2x^2 y}}{3 \cdot 2x^2 y} \\
&= \frac{\sqrt{24x^3 y}}{6x^2 y} \\
&= \frac{\sqrt{4x^2 \cdot 6xy}}{6x^2 y} \\
&= \frac{2x \cdot \sqrt{6xy}}{6x^2 y} \\
&= \frac{\sqrt{6xy}}{3xy}
\end{aligned}$$

4. Given that  $d > 0$ .

$$\begin{aligned}
\frac{cd}{\sqrt{cd^3}} &= \frac{cd\sqrt{cd^3}}{cd^3} \\
&= \frac{\sqrt{cd^3}}{d^2} \\
&= \frac{\sqrt{c} \cdot d^{3/2}}{d^2} \\
&= \frac{\sqrt{c} \cdot d^{1/2}}{d} \\
&= \frac{\sqrt{cd}}{d}
\end{aligned}$$

5. Given that  $p > 0$

$$\begin{aligned}\frac{7t}{\sqrt{14tp^2}} &= \frac{7t \cdot \sqrt{14tp^2}}{14tp^2} \\ &= \frac{7tp\sqrt{14t}}{14tp^2} \\ &= \frac{\sqrt{14t}}{2p}\end{aligned}$$

6. Given that  $a > 0$

$$\begin{aligned}\sqrt{\frac{5b}{a^3}} &= \frac{\sqrt{5b}}{\sqrt{a^3}} \\ &= \frac{\sqrt{5b}}{a\sqrt{a}} \\ &= \frac{\sqrt{5ab}}{a\sqrt{a^2}} \\ &= \frac{\sqrt{5ab}}{a^2}\end{aligned}$$

7.

$$\begin{aligned}\sqrt{\frac{14r}{p^5}} &= \frac{\sqrt{14r}}{\sqrt{p^5}} \\ &= \frac{\sqrt{14p^5r}}{p^5} \\ &= \frac{\sqrt{14r} \cdot p^{5/2}}{p^5} \\ &= \frac{\sqrt{14pr}}{p^3}\end{aligned}$$

8.

$$\begin{aligned}\sqrt{\frac{9a^2}{b^3}} &= \frac{\sqrt{9a^2}}{\sqrt{b^3}} \\ &= \frac{\sqrt{9a^2}}{b\sqrt{b}} \\ &= \frac{\sqrt{9a^2b}}{b\sqrt{b^2}} \\ &= \frac{3a\sqrt{b}}{b^2}\end{aligned}$$

9. Given that  $z > 0$ .

$$\begin{aligned}\sqrt{\frac{11}{wz^2}} &= \frac{\sqrt{11}}{\sqrt{wz^2}} \zeta = \frac{\sqrt{11}}{z\sqrt{w}} \\ &= \frac{\sqrt{11w}}{zw}\end{aligned}$$

10.

$$\begin{aligned}\sqrt{\frac{5a}{3b}} &= \frac{\sqrt{5a}}{\sqrt{3b}} \\ &= \frac{\sqrt{15ab}}{3b}\end{aligned}$$

11.

$$\begin{aligned}\sqrt{\frac{a^2+3}{a^2-3}} &= \frac{\sqrt{a^2+3}}{\sqrt{a^2-3}} \\ &= \frac{\sqrt{(a^2+3)(a^2-3)}}{a^2-3} \\ &= \frac{\sqrt{a^4-9}}{a^2-3} \\ &= \frac{\sqrt{a^4-9}}{a^2-3}\end{aligned}$$

12. Given  $y > 0$

$$\begin{aligned}\frac{2+3\sqrt{xy}}{3\sqrt{y}} &= \frac{(2+3\sqrt{xy}) \cdot \sqrt{y}}{3y} \\ &= \frac{3y\sqrt{x}+2\sqrt{y}}{3y} \\ &= \sqrt{x} + \frac{2\sqrt{y}}{3y}\end{aligned}$$

## Partial fraction decomposition

We have something of the form:

$$\frac{1}{(x-1)(3x-2)}$$

and we want to decompose it into partial fractions:

$$\frac{1}{(x-1)(3x-2)} = \frac{A}{x-1} + \frac{B}{3x-2}$$

Multiplying both sides by  $(x-1)(3x-2)$ , we get:

$$1 = A(3x-2) + B(x-1)$$

Setting  $x = 1$ , we get:

$$1 = A(3(1)-2) + B(1-1) = A$$

Setting  $x = \frac{2}{3}$ , we get:

$$1 = A\left(3\left(\frac{2}{3}\right)-2\right) + B\left(\frac{2}{3}-1\right) = -\frac{B}{3}$$

Solving for  $B$ , we get:

$$B = -3$$

So the partial fraction decomposition is:

$$\frac{1}{(x-1)(3x-2)} = \frac{1}{x-1} - \frac{3}{3x-2}$$

### Exercises

1. Decompose  $\frac{44x+48}{(x+6)(x-3)(x+2)}$

$$\frac{44x+48}{(x+6)(x-3)(x+2)} = \frac{A}{x+6} + \frac{B}{x-3} + \frac{C}{x+2}$$

$$44x+48 = A(x-3)(x+2) + B(x+6)(x+2) + C(x+6)(x-3)$$

Set  $x = 3$ :

$$44 \cdot 3 + 48 = B \cdot 9 \cdot 5$$

$$\frac{180}{9 \cdot 5} = B$$

$$B = 4$$

Set  $x = -2$ :

$$44 \cdot -2 + 48 = C \cdot 4 \cdot -5$$

$$-40 = C \cdot -20$$

$$C = 2$$

Set  $x = -6$ :

$$44 \cdot -6 + 48 = A \cdot -9 \cdot -4$$

$$A = -6$$

2. Given that

$$\frac{-2x-9}{x+3}x = \frac{1}{x+3} + \frac{B}{x}$$

find  $B$

$$\frac{-2x-9}{x+3}x = \frac{1}{x+3} + \frac{B}{x}$$

$$(-2x-9) = x + B(x+3)$$

$$-3x-9 = B(x+3)$$

$$\frac{-3(x+3)}{x+3} = B$$

$$B = -3$$

3. Given that

$$\frac{-9x+36}{(x-3)(x-6)} = -\frac{3}{x-3} + \frac{B}{x-6}$$

$$-9x+36 = -3(x-6) + B(x-3)$$

$$-6x+18 = B(x-3)$$

$$\frac{-6(x-3)}{x-3} = B$$

$$B = -6$$

## Functions

### Horizontal asymptotes of rational functions

Rational functions are of the form:

$$f(x) = \frac{ax^n + bx^{n-1} + \dots}{cx^m + dx^{m-1} + \dots}$$

To get the horizontal asymptotes, first take the **dominant term**. It's the leading term of the polynomial in the denominator, *without the coefficient*. Here  $x^m$ .

Then, divide every term in both numerator and denominator by the dominant term, and evaluate at  $x \rightarrow \infty$ .

### Exercises

- Determine the horizontal asymptote of  $f(x) = \frac{x^3+2x+1}{x^2-x-12}$

The dominant term in  $x^2$ . Let's divide by it.

$$\frac{\frac{x^3+2x+1}{x^2}}{\frac{x^2-x-12}{x^2}} = \frac{\left(x + \frac{2}{x} + \frac{1}{x^2}\right)}{1 - \frac{x+12}{x^2}}$$

As  $x \rightarrow \infty$ , this becomes  $\frac{x}{1} = x$ . Since this is not a constant, there is no horizontal asymptote.

- Find the horizontal asymptote of  $\frac{3x^2+3}{6x^2+x}$ .

The dominant term is  $x^2$ .

$$\frac{\frac{3x^2+3}{x^2}}{\frac{6x^2+x}{x^2}} = \frac{3 + \frac{3}{x^2}}{6 + \frac{1}{x}}$$

As  $x \rightarrow \infty$ , this becomes  $\frac{3}{6} = \frac{1}{2}$ . The horizontal asymptote is  $y = \frac{1}{2}$ .

- Find the horizontal asymptote of  $f(x) = \frac{3x^2-2}{2x^2+1}$ .

The dominant term is  $x^2$ .

$$\frac{\frac{3x^2-2}{x^2}}{\frac{2x^2+1}{x^2}} = \frac{3 - \frac{2}{x^2}}{2 + \frac{1}{x^2}}$$

As  $x \rightarrow \infty$ , this becomes  $\frac{3}{2}$ . The horizontal asymptote is  $y = \frac{3}{2}$ .

### Domain

All the values where the radicand  $> 0$  for even degrees,  $(-\infty, \infty)$  for odd degrees.

### Exercises

1. Domain of  $d(x) = \sqrt[7]{-2x + \frac{1}{2}} + 1$ ?

$$-2x + \frac{1}{2} >$$

### Roots of rational functions

First factor numerator and denominator, then cancel out common factors, then set the numerator to zero and solve.

### Exercises

1. Solve  $f(t) = 0$  where  $f(t) = \frac{7t^2 + 14}{t^2 - 25}$ .

$$\frac{7t^2 + 14}{t^2 - 25} = 0$$

$$\frac{7(t^2 + 2)}{(t + 5)(t - 5)} = 0$$

$$t^2 + 2 = 0$$

$$t^2 = -2$$

2. Solve  $f(x) = 0$  where  $f(x) = \frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45}$ .

$$\frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45} = 0$$

$$\frac{x^2 + 5x - 3x - 15}{x^2(x + 5) - 9(x + 5)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x^2 - 9)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x - 3)(x + 3)} = 0$$

$$1 = 0$$

The equation has no solution.

3. Solve  $f(t) = 0$  where  $f(t) = \frac{t-3}{t^2+6t+5}$

$$\frac{t-3}{t^2+6t+5} = 0$$

$$\frac{t-3}{t^2+t+5t+5} = 0$$

$$\frac{t-3}{(t+5)(t+1)} = 0$$

$$t-3 = 0$$

$$t = 3$$

## Invertible functions

A function is invertible on an interval if and only if it is one-to-one on that interval.

To check if a function is one-to-one, we can use the horizontal line test. If any horizontal line intersects the graph of the function at most once, then the function is one-to-one.

## Statistics

### Variance

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

### Covariance

Covariance is the measure of the joint variability of two random variables.

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

### Sum of squares

The sum of squares is the sum of the squared differences between each data point and the mean.



$$\sum_{i=1}^n (x_i - \bar{x})^2$$

Denoted  $S_{xx}$ . If two variables are involved (covariance), it is denoted as  $S_{xy}$ .

Shortcut formula:

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{1}{n(\sum_{i=1}^n x_i)^2}$$

$$S_{xy} = \sum_{i=1}^n x_i y_i - \frac{1}{n(\sum_{i=1}^n x_i)} \left( \sum_{i=1}^n y_i \right)$$

## Mean estimation for continuous grouped data

$$\bar{x} \approx \frac{1}{n} \sum_{i=1}^K f_i \cdot m_i$$

where

- $f_i$  is the frequency of the  $i$ th interval
- $m_i$  is the midpoint of the  $i$ th interval
- $n$  is the total number of observations

## Estimating a variance for continuous grouped data

Using the same notation as above:

$$\sigma_n^2 \approx \frac{1}{n} \sum_{i=1}^K f_i \cdot (m_i - \bar{x})^2$$

## Graphing

### Graphing cubic curves with one distinct real root

If we are given one root of a cubic equation, we can use synthetic division to factor it.

#### Example

Given that  $(x - 3)$  is a factor of  $f(x) = x^3 + x^2 - 4x - 24$ , sketch the curve of  $y = f(x)$ .

First, synthetic division:

|   | $x^3$ | $x^2$ | $x^1$ | $x^0$ |
|---|-------|-------|-------|-------|
| 3 | 1     | 1     | -4    | -24   |
|   |       | 3     | 12    | 24    |
|   | 1     | 4     | 8     | 0     |

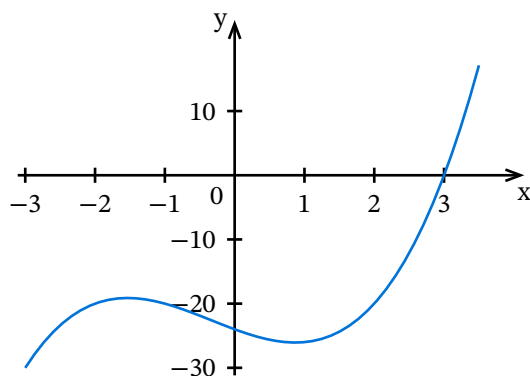
So we have  $f(x) = (x - 3)(x^2 + 4x + 8)$

$\therefore x = 3$  is a root.

The discriminant of  $x^2 + 4x + 8$  is  $4^2 - 4 \cdot 1 \cdot 8 = -16$ , which is negative, so  $x = 3$  is the only real root.

The  $y$  intercept is  $f(0) = 0^3 + 0^2 - 4 \cdot 0 - 24 = -24$

Now we can draw the curve:



## Logarithms

### Domain

$(0, \infty)$

### Range

$(-\infty, \infty)$

## Complex Numbers

### Magnitude

The magnitude of a complex number  $z = a + bi$  is defined as  $|z| = \sqrt{a^2 + b^2}$ .

### Argument

For any complex number, like  $z = 3 + 4i$ , there is  $\arg(z)$  that is the angle (in radians) between the  $x$  axis and the complex number in the Argand diagram.

So for  $z$ :

$$\begin{aligned}\tan \theta &= \frac{4}{3} \\ \theta &= \arctan\left(\frac{4}{3}\right) \\ &\approx 0.93\end{aligned}$$

## Trigonometry

### Unit circle

#### Reference angle

The *reference angle* of an angle is the smallest angle between the terminal side of the angle and the  $x$ -axis. It is always a positive angle less than or equal to 90 degrees.

#### Pythagorean identity in the first quadrant

In the first quadrant, any point  $(x, y)$  forms a right triangle with the origin, where  $x$  and  $y$  are the lengths of the legs and  $\sqrt{x^2 + y^2}$  is the length of the hypotenuse. Since in the unit circle, the length of the hypotenuse is 1, by definition, the Pythagorean identity states that  $x^2 + y^2 = 1$ .

### Exercises

1. For a point in the first quadrant with  $y = \frac{\sqrt{5}}{3}$ , find its  $x$  coordinate.

$$\sqrt{\left(\frac{\sqrt{5}}{3}\right)^2} + x^2 = 1$$

$$\left(\frac{\sqrt{5}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{5}}{3}\right)^2$$

$$x^2 = 1 - \frac{5}{9}$$

$$x^2 = \frac{4}{9}$$

$$x = \sqrt{\frac{4}{9}}$$

$$x = \frac{2}{3}$$

2. Point in the first quadrant.  $y$  coordinate is  $\frac{\sqrt{2}}{3}$ . What is the  $x$  coordinate?

$$\sqrt{\left(\frac{\sqrt{2}}{3}\right)^2} + x^2 = 1$$

$$\left(\frac{\sqrt{2}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{2}}{3}\right)^2$$

$$x^2 = 1 - \frac{2}{9}$$

$$x^2 = \frac{7}{9}$$

$$x = \sqrt{\frac{7}{9}}$$

$$x = \frac{\sqrt{7}}{3}$$

### Pythagorean trigonometric identity

In the unit circle, since for any central angle  $x = \cos \theta$  and  $y = \sin \theta$ .

That means

$$x^2 + y^2 = 1$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

This is the **Pythagorean trigonometric identity**. It works in all quadrants.

### Exercises

1. Point  $P$  on the unit circle with  $x = \frac{2}{5}$ . Find  $\sin \theta$ .

$$\begin{aligned}
 \sin^2 \theta + \cos^2 \theta &= 1 \\
 \sin^2 \theta &= 1 - \cos^2 \theta \\
 \sin \theta &= \sqrt{1 - \cos^2 \theta} \\
 \sin \theta &= \sqrt{1 - \frac{4}{25}} \\
 \sin \theta &= \sqrt{\frac{21}{25}} \\
 \sin \theta &= \frac{\sqrt{21}}{5}
 \end{aligned}$$

2.  $y$  coordinate is  $\frac{4}{9}$ . Find  $\cos \theta$ .

$$\begin{aligned}
 \sin^2 \theta + \cos^2 \theta &= 1 \\
 \left(\frac{4}{9}\right)^2 + \cos^2 \theta &= 1 \\
 \cos^2 \theta &= 1 - \left(\frac{4}{9}\right)^2 \\
 \cos^2 \theta &= 1 - \frac{16}{81} \\
 \cos^2 \theta &= \frac{65}{81} \\
 \cos \theta &= \sqrt{\frac{65}{81}} \\
 \cos \theta &= \frac{\sqrt{65}}{9}
 \end{aligned}$$

3.  $x$  coordinate is  $\frac{\sqrt{7}}{4}$ . Find  $\csc \theta$ .

$$\begin{aligned}
 \sin^2 \theta + \cos^2 \theta &= 1 \\
 \sin^2 \theta &= 1 - \cos^2 \theta \\
 \sin \theta &= \sqrt{1 - \cos^2 \theta} \\
 \sin \theta &= \sqrt{1 - \left(\frac{\sqrt{7}}{4}\right)^2} \\
 \sin \theta &= \sqrt{1 - \frac{7}{16}} \\
 \sin \theta &= \sqrt{\frac{9}{16}} \\
 \sin \theta &= \frac{3}{4} \\
 \therefore \\
 \csc \theta &= \frac{1}{\sin \theta} \\
 \csc \theta &= \frac{4}{3}
 \end{aligned}$$

4.  $x$  coordinate is  $\frac{3}{7}$ . Find  $\csc \theta$ .

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \frac{9}{49}}$$

$$\sin \theta = \sqrt{\frac{40}{49}}$$

$$\sin \theta = \frac{\sqrt{40}}{7}$$

$$\therefore$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{7}{\sqrt{40}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7\sqrt{10}}{20}$$

5.  $y$  coordinate is  $\frac{1}{9}$ . Find  $\tan \theta$ .

We have  $\sin \theta = \frac{1}{9}$ .

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$\cos \theta = \sqrt{1 - \left(\frac{1}{9}\right)^2}$$

$$\cos \theta = \sqrt{1 - \frac{1}{81}}$$

$$\cos \theta = \sqrt{\frac{80}{81}}$$

$$\cos \theta = \frac{\sqrt{80}}{9}$$

$$\cos \theta = \frac{4\sqrt{5}}{9}$$

$$\therefore$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \frac{\frac{1}{9}}{\frac{4\sqrt{5}}{9}}$$

$$\tan \theta = \frac{1}{9} \cdot \frac{9}{4\sqrt{5}}$$

$$\tan \theta = \frac{\sqrt{5}}{20}$$

6.  $x$  coordinate is  $\frac{\sqrt{11}}{6}$ . Find  $\cot(\theta)$

We have  $\cos \theta = \frac{\sqrt{11}}{6}$  and  $\cot(\theta) = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta} = \frac{\frac{\sqrt{11}}{6}}{\sin \theta}$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin^2 \theta = 1 - \left(\frac{\sqrt{11}}{6}\right)^2$$

$$\sin^2 \theta = 1 - \frac{11}{36}$$

$$\sin^2 \theta = \frac{25}{36}$$

$$\sin \theta = \sqrt{\frac{25}{36}}$$

$$\sin \theta = \frac{5}{6}$$

$\therefore$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cot \theta = \frac{\frac{\sqrt{11}}{6}}{\frac{5}{6}}$$

$$\cot \theta = \frac{\sqrt{11}}{5}$$

### Quadrantal angles

Quadrantal angles are angles that are multiples of  $90^\circ$ . For example,  $0^\circ$ ,  $90^\circ$ ,  $180^\circ$ ,  $270^\circ$ , and  $360^\circ$  are all quadrantal angles. They lie on an axis.

They are useful to find the values of trigonometric functions. Just remember that for points on the unit circle,  $\cos \theta = x$  and  $\sin \theta = y$ .

The corresponding angles on  $[0, 360^\circ]$  are  $0, \frac{\pi}{2}, \pi, 3\frac{\pi}{2}, 2\pi$ .

Example:  $\cos 0 + \sin\left(\frac{\pi}{2}\right) = 1 + 1 = 2$

### Examples

- Find the value of  $\tan(180^\circ) + \cot(270^\circ)$

$$\begin{aligned}\tan(180^\circ) + \cot(270^\circ) &= \frac{\sin(180^\circ)}{\cos(180^\circ)} + \frac{\cos(270^\circ)}{\sin(270^\circ)} \\ &= \frac{0}{-1} + \frac{0}{-1} \\ &= 0\end{aligned}$$

- $\sec\left(\frac{\pi}{2}\right)$ ?

$$\begin{aligned}\sec\left(\frac{\pi}{2}\right) &= \frac{1}{\cos\left(\frac{\pi}{2}\right)} \\ &= \frac{1}{0}\end{aligned}$$

So it's undefined.

## Law of sines

In any triangle, where  $a$ ,  $b$ , and  $c$  are the lengths of the sides opposite angles  $A$ ,  $B$ , and  $C$  respectively, the law of sines states that:

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

For acute angles, the formula can be flipped:

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

For obtuse angles, the same formula works, but it will always give acute angles. To find an obtuse angle, subtract the acute angle from  $180^\circ$ .

### Exercises

- Find  $a$  if  $A = 70^\circ$ ,  $B = 35^\circ$ ,  $b = 12$ .

$$\begin{aligned}\frac{a}{\sin(A)} &= \frac{b}{\sin(B)} \\ \frac{a}{\sin(70^\circ)} &= \frac{12}{\sin(35^\circ)} \\ a &= 12 \cdot \frac{\sin(70^\circ)}{\sin(35^\circ)} \\ a &\approx 12 \cdot \frac{0.9397}{0.5736} \\ a &\approx 19.66\end{aligned}$$

- $R$ ,  $Q$  and  $P$  form a triangle.  $PR = 21m$ .  $\angle Q = 89^\circ$ ,  $\angle P = 38^\circ$ . What is  $QR$ ?

$$\begin{aligned}\frac{QR}{\sin(\angle P)} &= \frac{PR}{\sin(\angle Q)} \\ \frac{QR}{\sin(38^\circ)} &= \frac{21}{\sin(89^\circ)} \\ QR &= \sin(38^\circ) \cdot \frac{21}{\sin(89^\circ)}\end{aligned}$$

- $\triangle XYZ$ .  $\angle Y = 57^\circ$ ,  $XZ = 35$ ,  $XY = 24$ . Find  $\angle Z$ .

$$\begin{aligned}\frac{\sin(\angle Z)}{XY} &= \frac{\sin(\angle Y)}{XZ} \\ \frac{\sin(\angle Z)}{24} &= \frac{\sin(57^\circ)}{35} \\ \sin(\angle Z) &= 24 \cdot \frac{\sin(57^\circ)}{35} \\ \sin(\angle Z) &\approx 0.575 \\ \angle Z &\approx 35^\circ\end{aligned}$$

## Calculus

### Limits

#### Limits of logarithmic functions

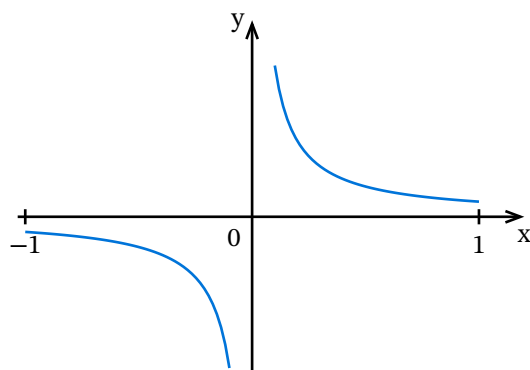
For any real  $n$ :

$$\lim_{x \rightarrow n} \log(x) = \log(x)$$

For infinity, remember the domain and apply transformations.

### Limits of reciprocal functions

Remember, reciprocal functions have the form  $f(x) = \frac{1}{x}$ . The limit of a reciprocal function as  $x$  approaches infinity is always zero.



In general, for any non-constant polynomial  $P(x)$ , we have  $\lim_{x \rightarrow \pm\infty} \frac{1}{P(x)} = 0$

But for  $\lim(x \rightarrow 0)$ , we have to consider the degree of the leading term of  $P(x)$ .

- Any function of the form  $y = \frac{1}{x^n}$  where  $n$  is odd has the same shape as  $y = \frac{1}{x}$ . There is no limit at  $x = 0$  because left and right limit are not the same.
- Any function of the form  $y = \frac{1}{x^n}$  where  $n$  is even has the same shape as  $y = \frac{1}{x^2}$ . There is a limit, since the function will approach  $\infty$  or  $-\infty$ .

## Differentiation

### Constant multiple rule

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x))$$

### Sum rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

For example for  $y = x^2 + x$ :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^2 + x) \\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(x) \\ &= 2x + 1 \end{aligned}$$

### The power rule

Given a monomial  $f(x) = x^n$ , where  $n$  is a real number,

$$f'(x) = nx^{n-1}$$

### Exercises

- Calculate  $\frac{dy}{dx}$  for  $y = 2x - \sqrt{x}$



$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}(2x - \sqrt{x}) \\
&= \frac{d}{dx}2x - \frac{d}{dx}\sqrt{x} \\
&= 2 - \frac{d}{dx}x^{1/2} \\
&= 2 - \frac{1}{2} \cdot x^{-1/2} \\
&= 2 - \frac{1}{2\sqrt{x}}
\end{aligned}$$

2. Find the derivative of  $f(x) = \frac{1}{8}x^8$

$$\begin{aligned}
f'(x) &= \frac{1}{8} \cdot \frac{d}{dx}x^8 \\
&= \frac{1}{8} \cdot 8x^7 \\
&= \frac{8x^7}{8} \\
&= x^7
\end{aligned}$$

3. Derivative of  $f(x) = 8\sqrt{x}$

$$\begin{aligned}
f'(x) &= \frac{d}{dx}8\sqrt{x} \\
&= \frac{d}{dx}8x^{1/2} \\
&= 8 \cdot \frac{d}{dx}\left(\frac{1}{2} \cdot x^{-1/2}\right) \\
&= 8 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}} \\
&= \frac{4}{\sqrt{x}}
\end{aligned}$$

### Rate of change

Average rate of change on an interval  $[a, a + h]$ :

$$\frac{\delta y}{\delta x} = \frac{f(a + h) - f(a)}{h}$$

The *instantaneous rate of change* at a point  $x = a$  is given by:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

### Exercises

1. Given that  $f(x) = x^2$ , what is  $f'(2)$  according to the definition?

$$\begin{aligned}
 f'(2) &= \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4h + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (4 + h) \\
 &= 4
 \end{aligned}$$

### Slope of a tangent line

The derivative of a function  $f(x)$  at a point  $x = a$  is the slope of the tangent line to the graph of  $f(x)$  at that point.

So to find the slope of the tangent line at a point  $(x_1, y_1)$ , we evaluate  $f'(x_1)$ .

### Exercises

- Find the slope of the tangent to  $y = 5x^7 - 3x^5 + 10x^3$  at the point where  $x = 1$ .

First find  $f'(x)$ :

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} (5x^7 - 3x^5 + 10x^3) \\
 f'(x) &= 5 \cdot \frac{d}{dx} x^7 - 3 \cdot \frac{d}{dx} x^5 + 10 \cdot \frac{d}{dx} x^3 \\
 f'(x) &= 5 \cdot 7x^6 - 3 \cdot 5x^4 + 10 \cdot 3x^2 \\
 f'(x) &= 35x^6 - 15x^4 + 30x^2 \\
 f'(x) &= 5(7x^6 - 3x^4 + 6x^2)
 \end{aligned}$$

- Find the slope of the tangent to  $y = 4x^3 + \frac{5}{2}x^2 - 200x$  at  $x = 4$ .

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} 4x^3 + \frac{5}{2}x^2 - 200x \\
 &= 12x^2 + 5x - 200
 \end{aligned}$$

The slope is  $f'(4) = 12$

- Find the  $x$ -coordinate of the point on the curve  $y = 2x^2 - 9x - 6$  whose tangent has a slope of 3.

First let's get the derivative of the function:

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} (2x^2 - 9x - 6) \\
 &= 4x - 9
 \end{aligned}$$

Now we set the derivative equal to the slope of the tangent line:

$$\begin{aligned}
 4x - 9 &= 3 \\
 4x &= 12 \\
 x &= 3
 \end{aligned}$$

## Integration

### The antiderivative

The antiderivative is the opposite of the derivative.

Infinitely many antiderivatives map to the same function.

For example, since the derivative of  $x^2$  is  $2x$ , we have  $x^2$  as an antiderivative of  $2x$ . But  $x^2 + 2$ ,  $x^2 + 1$ , etc. are also antiderivatives of  $2x$ .

In general, the antiderivatives of  $2x$  follow the pattern  $x^2 + C$  where  $C$  is a constant.

We can also formulate that using the  $\int$  symbol for integration:

$$\int 2x \, dx = x^2 + C$$

Where  $C$  is called the **constant of integration**.

These integrals are **indefinite integrals**.

The function being integrated (here  $2x$ ) is called the **integrand**.

### The power rule for integration

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

### Exercises

1. Antiderivative of  $\frac{1}{x^4}$ ?

$$\begin{aligned} \int \frac{1}{x^4} dx &= \int x^{-4} dx \\ &= \frac{x^{-4+1}}{-4+1} + C \\ &= \frac{x^{-3}}{-3} + C \\ &= -\frac{1}{3}x^3 + C \end{aligned}$$

2. Antiderivative of  $x^{\frac{5}{8}}$

$$\begin{aligned} \int x^{\frac{5}{8}} dx &= \frac{x^{\frac{5}{8}+1}}{\frac{5}{8}+1} + C \\ &= \frac{x^{\frac{13}{8}}}{\frac{13}{8}} + C \\ &= \frac{8x^{\frac{13}{8}}}{13} + C \end{aligned}$$

3. Antiderivative of  $\sqrt[5]{x^7}$

$$\begin{aligned} \int x^{\frac{7}{5}} dx &= \frac{x^{\frac{7}{5}+1}}{\frac{7}{5}+1} + C \\ &= \frac{x^{12/5}}{12/5} \\ &= \frac{5x^{12/5}}{12} \end{aligned}$$

4. Calculate  $\int \left(\frac{z}{5}\right) dz$

$$\begin{aligned}
 \int \left(\frac{z}{5} dz\right) &= \frac{1}{5} \int z dz \\
 &= \frac{1}{5} \cdot \left(\frac{z^2}{2} + C\right) \\
 &= \frac{z^2}{10} + C
 \end{aligned}$$

### The constant factor rule for indefinite integrals

To take the integral of a power function with a constant factor, like  $5x^2$ , we can take the constant factor out of the integral:

$$\int 5x^2 dx = 5 \cdot \int x^2 dx$$

This is the **constant factor rule**. In general:

$$\int k f(x) dx = k \int f(x) dx$$

Where  $k$  is a **constant**. It does not work if  $k$  is a variable.

### Exercises

1. Calculate  $\int \frac{1}{2y^5} dy$

$$\begin{aligned}
 \int \frac{1}{2y^5} dy &= \frac{1}{2} \int y^{-5} \\
 &= \frac{1}{2} \cdot \frac{y^{-5+1}}{-5+1} + C \\
 &= \frac{1}{2} \cdot \frac{y^{-4}}{-4} + C \\
 &= \frac{1}{2} \cdot -\frac{1}{4y^4} + C \\
 &= -\frac{1}{8y^4} + C
 \end{aligned}$$

2. Calculate  $\int \frac{1}{3x^6} dx$

$$\begin{aligned}
 \int \frac{1}{3x^6} dx &= \frac{1}{3} \cdot \int x^{-6} \\
 &= \frac{1}{3} \cdot \frac{x^{-5}}{-5} + C \\
 &= \frac{1}{3} \cdot -\frac{1}{5x^5} + C \\
 &= -\frac{1}{15x^5} + C
 \end{aligned}$$

3. Calculate  $\int \frac{\pi}{2} dz$

$$\begin{aligned}
 \int \frac{\pi}{2} dz &= \frac{\pi}{2} \int z^0 dz \\
 &= \frac{\pi}{2} \cdot z + C \\
 &= \frac{\pi z}{2} + C
 \end{aligned}$$

### The sum rule for indefinite integrals

To take the integral of a sum, you can take the sum of the integrals of each term:

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

The sum rule can be combined with the constant factor rule to integrate sums of constant multiples of functions:

$$\int (k_1 f(x) + k_2 g(x)) dx = k_1 \int f(x) dx + k_2 \int g(x) dx$$

### Exercises

- Calculate  $\int \sqrt{x}(2+x) dx$

$$\begin{aligned}\int \sqrt{x}(2+x) dx &= \int 2\sqrt{x} + x \cdot \sqrt{x} dx \\&= \int 2\sqrt{x} dx + \int x \cdot \sqrt{x} dx \\&= 2 \int \sqrt{x} dx + \int x^{3/2} dx \\&= 2 \cdot \frac{x^{3/2}}{\frac{3}{2}} + \frac{x^{5/2}}{\frac{5}{2}} + C \\&= \frac{4x^{3/2}}{3} + \frac{2x^{5/2}}{5} + C \\&= \frac{4\sqrt{x^3}}{3} + \frac{2\sqrt{x^5}}{5} + C\end{aligned}$$

- Calculate  $\int \frac{2+x^4}{x^2} dx$

$$\begin{aligned}\int \frac{2+x^4}{x^2} dx &= \int \frac{2}{x^2} dx + \int \frac{x^4}{x^2} dx \\&= \int 2 \cdot x^{-2} dx + \int x^2 dx \\&= 2 \cdot \int x^{-2} dx + \frac{x^3}{3} + C \\&= 2 \cdot -x^{-1} + \frac{x^3}{3} + C\end{aligned}$$

- Calculate:

$$\begin{aligned}\int (3-x^2)^2 dx &= \int (9-6x^2+x^4) dx \\&= 9 \int 1 dx - 6 \int x^2 dx + \int x^4 dx \\&= 9x - 6 \frac{x^3}{3} + \frac{x^5}{5} + C \\&= \frac{x^5}{5} - 2x^3 + 9x + C\end{aligned}$$

- Find the antiderivative of  $z^3 + 2z - 1$

$$\begin{aligned}
 \int z^3 + 2z - 1 \, dz &= \int z^3 \, dz + \int 2z \, dz - \int 1 \, dz \\
 &= \frac{z^4}{4} + 2\frac{z^2}{2} - z + C \\
 &= \frac{z^4}{4} + z^2 - z + C
 \end{aligned}$$